INDIAN INSTITUTE OF TECHNOLOGY JODHPUR
MAL3040: Probability and Statistics

Quiz 3 — Questions and Brief Answers

2026–27, Semester I • 24 October 2026 • 15 marks + 2 bonus

These are brief solutions for self-checking. Equivalent correct methods are acceptable.

1. Bias and variance of estimators [5 marks]

Suppose X_1 and X_2 are independent random variables satisfying $E[X_1] = E[X_2] = \mu$ and $\mathrm{Var}(X_1) = \mathrm{Var}(X_2) = \sigma^2 > 0$. Consider the following estimators of μ:

$$T_1 = \frac{X_1 + X_2}{2}, \qquad T_2 = \frac{X_1 + 3X_2}{4}.$$

(a) Determine whether T_1 and T_2 are unbiased estimators of μ. Justify your answers. [2]

(b) Calculate $\mathrm{Var}(T_1)$ and $\mathrm{Var}(T_2)$. [2]

(c) Which estimator would you prefer, and why? [1]

Brief answer. Both estimators are unbiased, since

$$E[T_1] = \frac{\mu + \mu}{2} = \mu, \qquad E[T_2] = \frac{\mu + 3\mu}{4} = \mu.$$

Using independence,

$$\mathrm{Var}(T_1) = \frac{\sigma^2 + \sigma^2}{4} = \frac{\sigma^2}{2}, \quad \mathrm{Var}(T_2) = \frac{\sigma^2 + 9\sigma^2}{16} = \frac{5\sigma^2}{8}.$$

Prefer T_1: both are unbiased, but T_1 has smaller variance.

2. Maximum likelihood estimation [5 marks]

The observations $2, 5, 3, 7$ are obtained independently from Uniform$(0, \theta)$, where $\theta > 0$ is unknown. The probability density function is

$$f(x;\theta) = \begin{cases} \dfrac{1}{\theta}, & 0 \le x \le \theta, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Write the likelihood function $L(\theta)$ for the given observations. [2]

(b) Find the maximum likelihood estimate of θ. Justify your answer. [2]

(c) Explain briefly why the maximum likelihood estimate cannot be smaller than the largest observation. [1]

Brief answer. The largest observation is 7, so

$$L(\theta) = \begin{cases} \theta^{-4}, & \theta \ge 7, \\ 0, & 0 < \theta < 7. \end{cases}$$

For $\theta \ge 7$, the likelihood θ^{-4} decreases as θ increases. Hence $\widehat{\theta} = 7$. Any $\theta < 7$ excludes an observed value and gives likelihood zero.

3. Confidence intervals [5 marks]

A random sample of size 16 from a normal population gives $\bar{x} = 20$ and $s = 4$. The population variance is unknown.

(a) Identify the distribution and degrees of freedom required for a two-sided 95% confidence interval for μ. Obtain the appropriate critical value from Ross's tables. [2]

(b) Construct the two-sided 95% confidence interval for μ. [2]

(c) Briefly explain what the 95% confidence level means in this context. [1]

Brief answer. Use Student's t distribution with 15 degrees of freedom. The upper-tail probability is 0.025, and the critical value is $t_{0.025,15} \approx 2.131$. Thus

$$\bar{x} \pm t_{0.025,15}\frac{s}{\sqrt{n}} = 20 \pm 2.131\frac{4}{\sqrt{16}} = \boxed{(17.869,\ 22.131)}.$$

Over repeated random samples, about 95% of intervals constructed by this procedure would contain the fixed population mean μ.

4. Bonus: Comparing variances of estimators [+2 marks]

Let $X_1, \ldots, X_n$ be independent observations from $N(\mu, \sigma^2)$, where $n \geq 2$. Consider the two estimators of σ^2:

$$T_1 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2, \qquad T_2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

You may use the fact that $\mathrm{Var}(U) = 2k$ when $U \sim \chi^2_k$.

(a) Find $\mathrm{Var}(T_1)$ and $\mathrm{Var}(T_2)$. [1]

(b) Which of these estimators has smaller variance? Does this alone tell you which is preferable? [1]

Brief answer. Since $Q = (n-1)T_2/\sigma^2 \sim \chi^2_{n-1}$,

$$\mathrm{Var}(T_2) = \frac{\sigma^4}{(n-1)^2}\,2(n-1) = \frac{2\sigma^4}{n-1}.$$

Also $T_1 = \frac{n-1}{n}T_2$, so

$$\mathrm{Var}(T_1) = \left(\frac{n-1}{n}\right)^2 \mathrm{Var}(T_2) = \frac{2(n-1)\sigma^4}{n^2}.$$

Thus T_1 has smaller variance for $\sigma^2 > 0$. This alone does not make it preferable: T_1 is biased, while T_2 is unbiased.

End of questions and brief answers